

Name: _____

TOPIC TEST

Trigonometry

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks
- Note: All diagrams are not to scale

Part A

10 multiple-choice questions

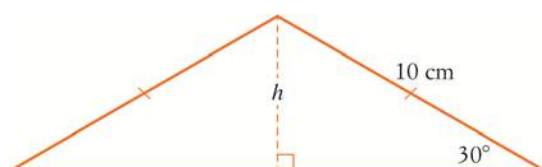
1 mark each: 10 marks

Circle the correct answer.

- 1 If $\tan \theta = \frac{m}{n}$, then which is the correct expression for $\frac{\cos \theta}{\sin \theta}$?

A $\frac{m}{n}$ B $\frac{n}{m}$
 C mn D $\frac{m^2}{n^2}$

- 2 Calculate h , the height of this isosceles triangle.

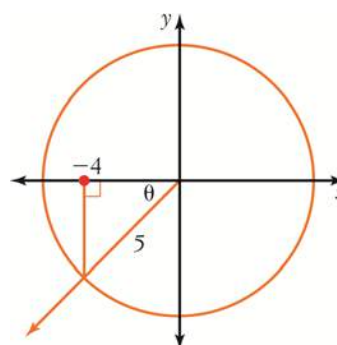


A 5 cm B 6 cm
 C 7.07 cm D 8.66 cm

- 3 A right-angled triangle with a 16.5 cm long hypotenuse has one angle of size $51^\circ 34'$. Find the length of its shortest side.

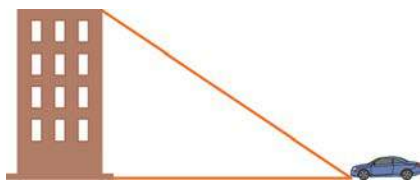
A 21.1 cm
 B 12.9 cm
 C 13.1 cm
 D 10.3 cm

- 4 Which trigonometric ratio below is correct for the triangle shown in the diagram?



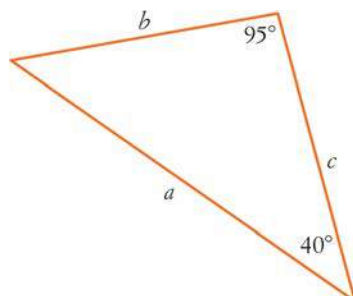
A $\sin \theta = \frac{4}{5}$ B $\cos \theta = \frac{4}{5}$
 C $\tan \theta = \frac{3}{4}$ D $\tan \theta = -\frac{3}{4}$

- 5 A car is parked 120 metres from the base of a building that is 235 metres tall. What is the angle of depression from the top of the building to the car?



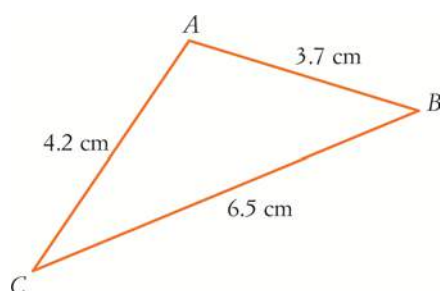
- A 63° B 59°
C 31° D 27°

- 6 Which is the correct sine rule for this triangle?



- A $\frac{a}{\sin 40^\circ} = \frac{b}{\sin 95^\circ}$ B $\frac{a}{\sin 95^\circ} = \frac{c}{\sin 40^\circ}$
C $\frac{b}{\sin 40^\circ} = \frac{c}{\sin 45^\circ}$ D $\frac{b}{\sin 40^\circ} = \frac{c}{\sin 95^\circ}$

- 7 What is the size of angle B in this triangle?



- A $32^\circ 12'$ B $37^\circ 23'$
C $37^\circ 14'$ D $110^\circ 34'$

- 8 Convert 80° to radians.

- A 80π B $\frac{\pi}{80}$
C $\frac{2\pi}{9}$ D $\frac{4\pi}{9}$

- 9 Find the exact value of $\tan\left(\frac{2\pi}{3}\right)$.

- A $-\frac{1}{\sqrt{3}}$ B $-\sqrt{3}$
C $\sqrt{3}$ D $\frac{1}{\sqrt{3}}$

- 10 Find the exact area of a sector with radius 4 cm and subtended angle 150° .

- A 20π B $\frac{20\pi}{3}$
C $\frac{10\pi}{3}$ D $\frac{5\pi}{3}$

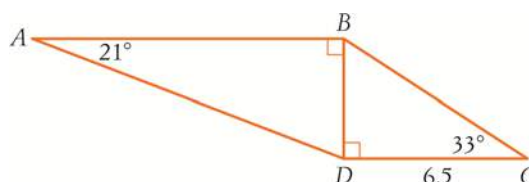
Part B

10 free-response questions

40 marks

Show your working where appropriate.

- 11 Show that the value of $AD = \frac{6.5 \tan 30^\circ}{\sin 21^\circ}$.

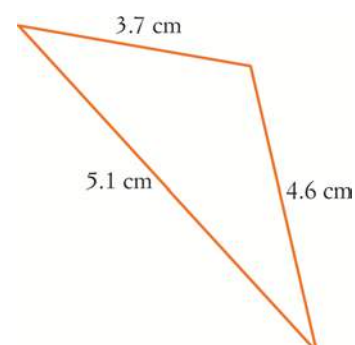


[4 marks]

- 12 A taxi drives due north from a hotel for 3 km and then turns to drive due west for 8 km to drop off a passenger at a conference centre. What is the bearing of the conference centre from the hotel, to the nearest degree?

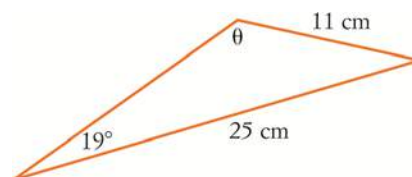
[4 marks]

- 13 Find the size of the smallest angle of this triangle, correct to the nearest degree.



[4 marks]

- 14** Calculate the value of the obtuse angle θ , correct to the nearest minute.



[4 marks]

- 15** A runner sets out on an orienteering course on a bearing of 160° from the starting base for 2.4 km and reaches his first control point. He then turns on a bearing of 225° and runs for 3.5 km to the second control point. From this point, he returns straight back to the starting base to complete the course.

a Sketch a diagram to display this information.

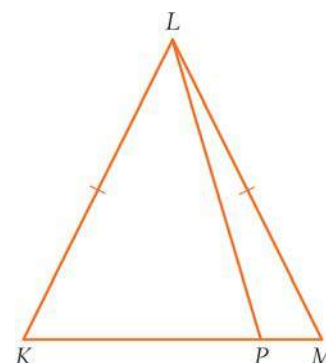
- b** Calculate the total distance the runner has completed around the course from the start to the finish, correct to one decimal place.

[6 marks]

- 16** In the diagram $KL = LM = 25$ cm.

$$\angle KLP = 70^\circ \text{ and } \angle MLP = 10^\circ.$$

What is the length of KP , correct to the nearest centimetre?



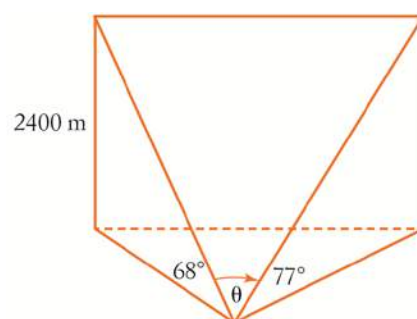
[4 marks]

- 17** Write the exact value of $\cos\left(\frac{\pi}{4}\right)$.

[2 marks]

- 18** A helicopter flying at 60 km/h at a constant altitude of 2400 m is observed to have an angle of elevation of 77° . After one minute, the angle of elevation is 68° .

Calculate the angle θ through which the observer has turned during the one minute.



[4 marks]

- 19** The area of a circle is 354 cm^2 . Find, in degrees and minutes, the angle subtended at the centre of the circle by a 2.9 cm arc.

[4 marks]

- 20** In $\triangle ABC$, $AB = 25 \text{ cm}$, $BC = 39 \text{ cm}$ and $\angle C = 38^\circ 4'$.

a Sketch a diagram to display this information.

- b** Find the size of the obtuse angle A , correct to the nearest minute.

[4 marks]

This is the end of the test.

Answers**1** B **2** A **3** D **4** C **5** A**6** C **7** C **8** D **9** B **10** B

11 $\tan 33^\circ = \frac{BD}{6.5}$

$$\therefore BD = 6.5 \tan 33^\circ$$

$$\sin 21^\circ = \frac{BD}{AD}$$

$$\text{i.e. } AD = \frac{6.5 \tan 33^\circ}{\sin 21^\circ}$$

12 291°

13 44°

14 $47^\circ 44'$

15 10.6 km

16 27 cm

17 $\frac{1}{\sqrt{2}}$

18 $22^\circ 40'$

19 $15^\circ 39'$

20 $105^\circ 52'$